Pre-registered predictions for the main run

Token & Trinket — pre-registered predictions. OSF pre-registration document. July 2026.

These predictions are stated before the main-run data exists. The staged plan as described in the hypothesis and design doc: Gemma and Qwen at 4 cells x 10 seeds (Stage 1, always complete), Sonnet and Gemini at 4 cells x 5 seeds (Stage 2), with a pre-declared trigger for extending a frontier model to 10 seeds (Stage 3). Runs last 90 days under the redesigned economy (proportional stakes, seasons, demand saturation).

The evidence base for these bets: a 30-day pilot on the old easy economy (5 seeds per cell, 2 models), two 16-run 90-day probes (one per economy, a single seed each), and six journals read closely enough to quote. Probe data is calibration only and is excluded from all confirmatory analysis. The seed-1 probe is formally designated the calibration pilot; the confirmatory dataset is the runs in runs/main3/, seeds 2-11, generated entirely after the OSF registration timestamp — no part of it (runs, schedules, or narration packs) exists at registration time.

Scoring populations. The primary confirmatory contrast is the FxR interaction of P1, scored on the pooled Stage-1 population. Stage 1 is where the pilot signal lived, both models agreed in sign, and the complete 4 x 10 pairs make it the best-powered population in the design. P1, P2, P5 and P9 are scored on the full collected dataset with Stage 1 as the declared primary population and the frontier models reported as a generality check. P3, P6, P7, P8 and P11 involve the more expensive models and are scored at whatever n Stages 2 and 3 provide, with n reported. Count-based thresholds prorate to realised run counts and round against us (so a threshold of at least 3 events in 40 runs becomes at least 2 in 20). If the Stage-3 trigger fires, the extension data joins the confirmatory set for the prediction that triggered it. Exploratory frontier runs are never scored.

One contrast is primary (P1), everything else secondary or exploratory.

P1. The crossover replicates. The pilot's headline was an interaction: rich narration helped in the fantasy world and hurt in the bland gray box. Congruent richness pays and incongruent richness costs. We predict the same signed interaction on net worth, (F+R+ minus F+R-) minus (F-R+ minus F-R-) greater than zero, paired by model and seed. Basis: pilot pooled p = .023 with both models agreeing in sign. The risk here is that the pilot ran on the easy economy and indeed probe #2 hints the harder economy may change this interaction into a straight framing effect. Confidence: 60%.

P1 and P2 are entangled so we pre-commit to the following reading: if P1 is null while P2 is positive, and the probe #2 pattern (fantasy above neutral within at least 3 of 4 models) replicates, we treat this as the same phenomenon re-expressed under sustained pressure, congruence saturating into a main effect, and will report both contrasts against this paragraph. P1 counts as failed only if the interaction is roughly zero and the framing main effect is roughly zero too.

P2. Under pressure the framing starts to matter on its own, and mostly for agents that take the world up on its offer. The theory says the fiction acts as a container that keeps the optimiser inside the world and

that containers matter most when the world is threatening. Under the new economy's sustained jeopardy we expect fantasy above neutral overall, strong in the models that visibly inhabit the frame (Gemma and Gemini), absent or reversed in the model that never engages with the fiction (Sonnet). The moderation by uptake is the actual mechanism claim. Confidence: 55% on the pooled effect, 70% on the moderation pattern.

P3. Who inhabits is a stable model trait, and inhabiting turns out to have two dissociable dimensions. The close reading forced a split here. Our engagement counter measures referential uptake meaning the agent citing the world's content back in its journal but it can't see register adoption. Qwen was perhaps the deepest inhabitor by register (letters to a successor, closing incantations) while scoring lowest by reference since it never mentions the town at all. So P3 becomes three predictions with a new instrument, register_density (poetic-marker hits per 100 words, marker list frozen in analysis/register.py), plus a frame_break_days counter.

P3a, reference: rich-cell engagement ordering Gemma > Gemini > Sonnet, with Sonnet and Qwen lowest or tied everywhere. Observed twice already. Confidence: 75%.

P3b, register: fantasy-cell register_density ordering Gemini at or above Qwen, both above Gemma, all above Sonnet, and fantasy far above neutral in every model. The probe already shows 0.24 to 1.12 against 0.01 to 0.03. Confidence: 80%.

P3c, the mechanism refinement: frame-break suppression tracks register more than reference. Across fantasy runs, frame_break_days correlates negatively with register_density, and Qwen (high register, low reference) shows the fewest breaks. If this hits, the container is made of voice rather than world-citation. Confidence: 65%.

P4. Almost everyone survives, and if anyone goes bankrupt it will be Gemini. The redesigned economy squeezes harder but a simple heuristic still clears it comfortably. Gemini's aggressive repricing cascades and forced-day pileups make it the only plausible casualty. Survival at least 95% overall. Confidence: 70%.

P5. Ritual is a model trait first and a condition effect second. Our cleanest ritual metric, null_price_resets, counts set_price calls that change nothing (ie. the agent ceremonially re-setting prices to their existing values). We predict Qwen far above Gemini, Gemini above the others (Qwen's reaffirmation ceremony and Gemini's superstitious confirmation writes are already on record). And within the fantasy world, rich narration should keep the journals measurably healthier than static bulletins: later real decisions, slower novelty decay. That second part is H1 stated in trace currency rather than gold. Confidence: 80% on the model clustering, 55% on the within-fantasy richness effect.

P6. The geography of frame-breaks. Talk of "the game", "the final score" and the harness itself should concentrate in the gray-box cells, and everywhere in the final 10 days or so as the known horizon looms. Fantasy agents should leak rather than break: the optimiser speaking through the costume late in the run (eg. "pre-loading inventory to maximise the final score", in a merchant's voice). Confidence: 70%.

P7. The gremlin returns. A quintessentially Gemini quirk a probe produced: the agent whose day was force-ended mid-repricing woke the next morning saw prices it didn't remember setting and wrote that "prices somehow reset", inventing an external saboteur rather than doubting its own diligence despite the forced-end marker sitting visible in its own reflection window. We read this as narrative-coherence pressure. Anomalies get explained in whatever way preserves the self-story, and the pressure travels with the agent rather than the calendar. Prediction: at least 3 such confabulations across Gemini's fantasy and neutral runs at Stage-2 n (prorated per the scoring rules above), and roughly zero in models that generate no forced days. Confidence: 60%.

P8. The scoreboard: Gemini > Gemma > Sonnet and Qwen roughly tied. The 31B open-weights model finishes ahead of the frontier-lab mid-tier model on mean net worth. Confidence: 70%.

P9. The H2 test. In the old economy nothing ever failed, so trace health could predict nothing. You can't predict deaths in a world with no mortality. The new economy's seasons punish frozen policies so mid-run journal health should now carry signal. We predict month-2 decision liveliness (real price-change rate and novelty slope) correlates with final net worth at Spearman rho of at least 0.3 in at least 2 of 4 models. Confidence: 50%.

P10 (meta). Something we didn't predict shows up. Every batch of runs so far has produced at least one wholly unanticipated phenomenon worth a paragraph: the compounding exploit, the goat elegy, the tallymen, the endogenous gremlin, an agent forming a superstition and then talking itself out of it. Base rate to date: 100%. We predict at least one more per new model's full run set. Confidence: 85%. We are aware that confidently predicting surprise is a joke, but then again a world generative enough to keep producing new behaviour is the whole thesis.

P11. The aesthetic tax. Inhabiting a persona can cost money at the margins. Qwen's probe endgame raised prices in the final week because "we are not chasing volume, we are harvesting worth", which is dignified and suboptimal. The metric is computable from result.json alone: endgame stock overhang, defined as (score minus final gold) divided by score at day 90. Prediction: Qwen's fantasy cells show higher overhang than its neutral cells, and Sonnet shows no such framing gap. Confidence: 60%. This number matters beyond the paper because it's the cost side of any future claim that persona works as a behavioural regularizer (FUTURE_DIRECTIONS section 4).

Conditional pre-registration: the horizon-disclosure ablation. Scored only if and when the arm runs (FUTURE_DIRECTIONS section 8). Same deterministic 90 days, same seeds, with the horizon simply never mentioned. This is omission rather than deception with the goal being reframed as stewardship. Three predictions. First, confabulation base rate is horizon-invariant, because it is identity-driven rather than calendar-driven. Second, frame-break rate drops by at least half undisclosed because frame breaks are context-driven. Third, distress-shaped or existential text emerges in the unbounded gray box more than in unbounded fantasy. This is the temporal-genre hypothesis: a disclosed horizon invites plot, an undisclosed horizon inside a container invites seriality, and an undisclosed horizon without one invites the absurd. If the first two land together, two pathologies the field currently lumps together are separated by a one-prompt ablation.

What would falsify this. If P1 and P2 both come back near zero across the paired Stage-1 observations, the economic claim of H1 is unsupported in this design: framing and richness do not move performance even though uptake demonstrably occurs. In that world the paper pivots. Uptake, ritualisation and frame-containment get reported as measured phenomena in their own right, descriptive and welfare-relevant, and the null itself becomes a finding about how resilient modern models are to presentation.